



Intensive ROS2 Course

Final Challenge

The objective of this mini challenge is to design, configure, and validate a **Digital Twin** of a real robot using **ROS2**. The goal is to achieve a virtual replica that mimics the geometry and kinematics of the physical system.

The challenge may be solved in teams, but all video reports, short reels and certification payments **MUST** be done individually.

The challenge consists of two parts:

- Video Report
- Short Reel

Deadline: October 2nd, 2026

Part 1: Video Report

Video Duration: 3 - 5 minutes.

Required Sections	%
Introduction Briefly explain the physical system that you will replicate in the virtual environment. This must include: • Robot name and type (e.g., industrial robotic arm, AGV/rover, drone, etc.) • Number of degrees of freedom (DoF) and general dimensions • Application for which the robot was designed.	10
Kinematic & Geometric Modeling • Links, joints, movement limits, and visual geometries correctly represent the selected robot.	15
Physics & Environment Simulation • The robot materializes stably in the simulated world without drifting, vibrating, or glitching.	15
ROS2 Control & Interfaces • Correct integration of essential publisher/subscriber nodes. • The digital twin successfully receives motion commands and publishes its joint_states.	15



Visualization & Monitoring • Clear, real-time visualization of the robot's state in RViz2. • Usage of static and dynamic transforms (TFs) and joint state changes.	15
Launch Files & Documentation • A single cleanly configured ROS2 launch file initiates the entire environment (robot state publisher, simulator, and RViz). • Clean repository organization with a short README explaining execution steps.	15
Conclusions A brief set of conclusions from the challenge and learning outcomes	15

Part 2: Short Reel

Video Duration: 50 – 60 seconds

Submission Details: You MUST deliver a short video Reel demonstrating your working digital twin. The reel must be published in Instagram and you must add **@manchester_robotics** as collaborators.

Specification: Start the video by briefly introducing the real-world robot you chose to model, including its name, degrees of freedom, and primary application. Next, show your digital twin in action: screen-record a single command executing your launch file, show the robot spawning stably into the simulator. Make it engaging!

Registration and payment

To be considered for certification you must complete the \$1500 MXN payment and submit your work and invoice by October 2nd, 2026.

- Invoice must be sent to: carmen.zazueta@manchester-robotics.com
- Challenge must be uploaded to: <https://forms.cloud.microsoft/e/nSwWce53eC>

Transfer payment details:

Name: Carmen Lucía Zazueta Esquer

Bank: Santander

CLABE: 014767110006049640

Description: ROS2 Your Name